

Efficacy of Musculoskeletal Manipulation for Endoscopic Discomfort in Patients Undergoing Upper Gastrointestinal Endoscopy and Colonoscopy: A Systematic Review and Meta-Analysis

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Abstract

Patients undergoing upper gastrointestinal endoscopy (UGIE) and colonoscopy often experience discomfort, including anxiety, abdominal pain, and bloating. The efficacy of musculoskeletal manipulation (e.g., massage) for these symptoms remains unclear. This systematic review and meta-analysis aimed to clarify its efficacy in patients undergoing UGIE and colonoscopy. We searched major electronic databases (MEDLINE (PubMed), CENTRAL, EMBASE, CINAHL, and PEDro) and trial registries up to July 17, 2025, for randomized controlled trials (RCTs). Meta-analyses employed a random-effects model, and risk of bias was formally assessed. Four trials were included. Two trials on reflexology showed a non-significant effect on pre-procedural anxiety, with the evidence being highly variable and leading to very low certainty findings. Two trials on abdominal massage following colonoscopy suggested a possible benefit for reduced abdominal pain, but this finding was limited by substantial heterogeneity. The overall evidence for musculoskeletal manipulation on anxiety, abdominal pain, and bloating is generally low or very low certainty due to imprecision and high risk of bias. While the overall evidence is insufficient for routine clinical recommendations, future high-quality, large-scale trials that standardize outcome assessments are essential to establish definitive conclusions. This review was registered with PROSPERO (CRD420251104522).

1. Introduction

Upper gastrointestinal endoscopy (UGIE) and colonoscopy are core diagnostic and therapeutic procedures in gastroenterology, yet a substantial proportion of patients report considerable peri-procedural discomfort. Commonly reported symptoms include anticipatory anxiety, abdominal pain related to insufflation, and post-procedural bloating. Non-pharmacological adjuncts — including reflexology, therapeutic massage, and other forms of musculoskeletal manipulation — have been proposed as low-risk interventions to reduce this burden without the sedation-related risks associated with pharmacological approaches. However, the evidence base for these interventions has not previously been synthesized in a systematic manner.

The objective of this review was to systematically identify, appraise, and synthesize randomized controlled trials evaluating musculoskeletal manipulation for procedural discomfort in patients undergoing UGIE or colonoscopy, and to grade the certainty of the resulting evidence using standard GRADE methodology.

2. Methods

2.1 Protocol and registration

This review was conducted in accordance with the PRISMA 2020 statement and was prospectively registered with PROSPERO (CRD420251104522). No amendments were made to the registered protocol.

2.2 Search strategy

We searched MEDLINE (via PubMed), CENTRAL, EMBASE, CINAHL, and PEDro from inception to July 17, 2025, without language restriction. Trial registries (ClinicalTrials.gov and ICTRP) were additionally searched to identify unpublished or ongoing trials. Reference lists of included studies and relevant reviews were hand-searched for additional eligible trials.

2.3 Eligibility criteria

Eligible studies were randomized controlled trials enrolling adult patients scheduled for UGIE and/or colonoscopy, comparing any form of musculoskeletal manipulation (massage, reflexology, or related manual therapy) against usual care, sham intervention, or an active comparator. Case reports, narrative reviews, and study protocols without outcome data were excluded.

2.4 Risk of bias and certainty assessment

Two reviewers independently assessed risk of bias using the Cochrane RoB 2 tool and resolved disagreements by consensus. The overall certainty of evidence for each outcome was rated using the GRADE framework (high / moderate / low / very low).

3. Results

3.1 Study selection

The search identified 812 records after deduplication. Following title/abstract screening, 41 reports underwent full-text assessment, of which four randomized controlled trials met all eligibility criteria and were included in the qualitative and quantitative synthesis.

Study	Design	N	Intervention	Primary outcome
Trial A	Parallel RCT	64	Foot reflexology	Pre-procedural anxiety (VAS)
Trial B	Parallel RCT	58	Auricular reflexology	Pre-procedural anxiety (STAI)
Trial C	Parallel RCT	72	Abdominal massage	Post-procedural abdominal pain
Trial D	Crossover RCT	45	Abdominal massage	Bloating score at 2 hours

3.2 Effect on pre-procedural anxiety

Pooling of the two reflexology trials showed a non-significant reduction in anxiety scores (standardized mean difference -0.18 , 95% CI -0.46 to 0.10 , $I^2 = 42\%$), with the certainty of evidence rated very low owing to serious risk of bias and imprecision.

3.3 Effect on abdominal pain and bloating

The two massage trials suggested a possible reduction in post-procedural abdominal pain (mean difference -0.9 points on a 10-point scale, 95% CI -1.8 to 0.0), although substantial heterogeneity ($I^2 = 61\%$) limited the precision of this pooled estimate. Effects on bloating were inconsistent across the two contributing trials.

3.4 Adverse events

No serious adverse events attributable to the interventions were reported in any of the four included trials.

4. Discussion

This systematic review found only four eligible randomized controlled trials evaluating musculoskeletal manipulation for endoscopy-related discomfort, and the overall certainty of the evidence was low to very low across all assessed outcomes. While a possible benefit for post-procedural abdominal pain was observed, this finding should be interpreted cautiously given the small number of trials, modest sample sizes, and substantial statistical heterogeneity.

Several factors likely contribute to the low certainty of the current evidence base, including inconsistent outcome measurement instruments across trials, limited blinding of outcome assessors, and heterogeneous intervention protocols (e.g., differing massage duration and technique). Future trials should adopt standardized, validated outcome measures and clearly reported intervention protocols to enable more precise pooled estimates.

4.1 Limitations

Limitations of this review include the small number of eligible trials, the exclusion of non-English language studies published only as conference abstracts, and the inability to formally assess publication bias given the limited number of included studies.

5. Conclusion

The current evidence for musculoskeletal manipulation as an adjunct to reduce procedural discomfort during upper gastrointestinal endoscopy and colonoscopy is of low to very low certainty. Although a possible benefit on post-procedural abdominal pain was suggested, routine clinical adoption cannot yet be recommended. Adequately powered, methodologically rigorous trials with standardized outcome reporting are needed before definitive conclusions can be drawn.

References

1. Demo Fixture Group. Reflexology for procedural anxiety in endoscopy: a randomized controlled trial. *J Demo Endosc.* 2024;12(3):101-109.
2. Demo Fixture Group. Auricular reflexology before upper GI endoscopy: a parallel-group RCT. *Demo J Gastrointest Nurs.* 2023;9(2):55-63.
3. Demo Fixture Group. Abdominal massage after colonoscopy reduces short-term pain: a randomized trial. *Demo Ann Colonoscopy.* 2022;5(1):12-19.
4. Demo Fixture Group. Effect of abdominal massage on post-colonoscopy bloating: a crossover trial. *Demo Clin Endosc Rep.* 2021;8(4):200-207.

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